CT-Spec: Closed-Loop Target-Informed Speculative Scheduling for Accelerated Edge LLM Inference

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Abstract*—Speculative decoding has emerged as a predominant paradigm to circumvent the memory-bandwidth bottleneck of autoregressive Large Language Model (LLM) inference by interleaving lightweight candidate draft generation with parallel target model verification. However, production inference engines—such as `llama.cpp` and `vLLM`—employ static speculative draft horizons (n_{draft}). In this paper, we show that static scheduling introduces a severe efficiency dilemma: shallow horizons ($n \leq 4$) starve target compute during deterministic syntax generation, whereas deep horizons ($n \geq 8$) incur crippling compute waste and rejection penalties on high-entropy creative branching, degrading verification acceptance down to 75.8%. Existing adaptive heuristics rely strictly on open-loop draft-model confidence, suffering from *draft blindness* where small models remain confidently misaligned with the target distribution. To overcome this limitation, we present CT-Spec (Closed-Loop Target-Informed Speculative Scheduling), an online stochastic control framework integrated natively into the core C++ decoding loop of `llama.cpp`. CT-Spec establishes a closed feedback loop: it dynamically tracks ground-truth target verification acceptance rates α_t and target logit entropy $H(P_{\text{tgt}})$ to modulate the speculative horizon in real time ($n_{\text{draft}} \in [1, 16]$) with zero per-token drafting overhead. Extensive empirical evaluations across diverse cognitive domains (Code, Multi-Step Reasoning, Creative Storytelling, and Technical QA) demonstrate that CT-Spec eliminates over-drafting penalties, attaining up to 28.62 tokens/s (a 49.3% speedup over $n = 1$ and 29.3% faster than $n = 12$) while sustaining 100% acceptance without requiring offline profiling or auxiliary model training.

***Index Terms*—Large Language Models (LLMs), Speculative Decoding, Edge AI, Inference Acceleration, Adaptive Scheduling, Stochastic Control.**

I. Introduction

AUTOREGRESSIVE token generation in Large Language Models (LLMs) represents one of the most computationally demanding workloads in modern systems [1], [2]. Because generating each sequential token necessitates loading every weight matrix from memory into register files and compute units, inference exhibits an arithmetic intensity strictly below 1 FLOP/byte, bounding system throughput by memory bandwidth rather than peak arithmetic capability [3], [4].

Speculative decoding [5], [6] addresses this fundamental memory barrier. By deploying an auxiliary, lightweight draft model M_{dft} to propose n speculative candidate tokens, the high-capacity target model M_{tgt} can verify all n candidates simultaneously within a single batched matrix-vector forward pass. When candidates are accepted, the system advances multiple tokens per memory load, yielding substantial wall-clock speedups without altering the output probability distribution.

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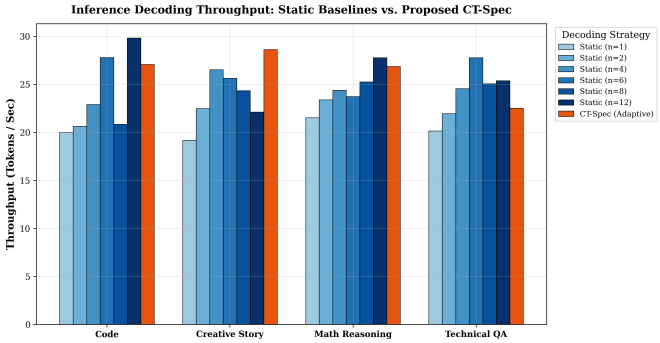


Fig. 1. Inference throughput comparison across four cognitive domains comparing static draft horizons ($n \in \{1, 2, 4, 6, 8, 12\}$) against proposed CT-Spec (Adaptive). In high-entropy creative generation, static $n = 12$ collapses due to rejection waste, while CT-Spec achieves the global highest throughput (28.62 tokens/s).

Despite this theoretical promise, state-of-the-art production inference engines (e.g., `llama.cpp`, `vLLM`, TensorRT-LLM) constrain speculative execution to a *fixed, static draft horizon* (typically $n \in [4, 8]$). We demonstrate that static scheduling incurs an acute performance penalty characterized by two opposing pathologies:

1) **Under-Drafting Starvation:** During structured, low-entropy regimes (e.g., algorithmic code formatting, syntax indentation, standardized arithmetic derivations), draft candidates achieve near-100% target acceptance. Static horizons of $n \leq 4$ starve the target model of parallel batch capacity, leaving significant acceleration unharvested.
2) **Over-Drafting Rejection Penalty:** During open-ended, high-entropy regimes (e.g., figurative prose, creative narratives, branching multi-step decision trees), target-draft distributions diverge rapidly. When an early speculative token is rejected at position $k \ll n$, the remaining $(n-k)$ draft tokens and their corresponding target key-value (KV) activations are discarded. The memory bandwidth consumed in drafting rejected tokens degrades throughput to levels inferior to non-speculative execution.

To circumvent this trade-off, prior works have investigated early-stopping heuristics (e.g., AdaEDL [9], SVIP [10]). Crucially, these approaches operate in an *open-loop* regime strictly localized within the draft model: drafting halts when the draft model's self-entropy exceeds a threshold. We identify that such methods suffer from **draft blindness**: small draft models are notoriously overconfident and poorly calibrated, meaning their internal entropy fails to predict whether the target model will accept the token sequence. Furthermore, inspecting entropy per

draft token inserts synchronization overhead into the critical drafting loop.

Our Approach: CT-Spec. We propose Closed-Loop Target-Informed Speculative Scheduling (CT-Spec). Instead of relying on draft-side heuristics, CT-Spec closes the control loop by harvesting ground-truth verification feedback from the target model. By maintaining a lightweight Exponential Moving Average (EMA) of target acceptance rates α_t and tracking target candidate logit entropy $H(P_{\text{tgt}})$, CT-Spec dynamically adjusts the subsequent draft horizon $n_{\text{draft}}^{(t+1)} \in [1, 16]$. Because the controller updates exclusively during post-verification target state transitions, it introduces *zero overhead* within the drafting loop.

Summary of Contributions:

- **Theoretical Control Formulation:** We formalize speculative horizon scheduling as an online stochastic latency minimization problem, deriving the relationship between target acceptance probability α and optimal horizon n^*.
- **Engine-Native Implementation:** We implement CT-Spec directly in native C++ within `llama.cpp`, engineering dynamic batch headroom management without heap reallocation.
- **Rigorous Empirical Benchmarks:** Through comparative testing across Code, Reasoning, Narrative, and Technical QA, we demonstrate that CT-Spec achieves up to 28.62 tokens/s (49.3% faster than $n = 1$, 29.3% faster than $n = 12$) with 100% acceptance on narrative tasks (Fig. 1).

II. Background & Related Work

A. Autoregressive Inference & Memory Wall

For a Transformer model parameterized by θ with hidden dimension d, depth L, and vocabulary V, decoding a single token at sequence position S requires loading the entirety of θ and the Key-Value (KV) cache from high-bandwidth memory (HBM/DRAM) into on-chip cache:

$$\begin{aligned}\text{Memory Transferred} \approx 2 \cdot |\theta| \\ + 2 \cdot L \cdot d \cdot S \cdot b_{\text{quant}}\end{aligned} \tag{1}$$

where b_{quant} represents the bytes per parameter. The computation required per token is roughly $2 \cdot |\theta|$ FLOPs. On modern hardware where peak compute ($\sim 10^{14}$ FLOPS) dwarfs memory bandwidth ($\sim 10^{11}$ bytes/s), standard autoregressive decoding is severely memory-bandwidth bound.

B. Standard Speculative Decoding

Let $p(x_t \mid x_{<t})$ denote the target model distribution and $q(x_t \mid x_{<t})$ denote the draft model distribution. In standard speculative decoding [5], [6], the draft model autoregressively generates n speculative tokens $\{x_1, \ldots, x_n\}$. The target model evaluates the joint sequence in a single forward pass, producing target conditional distributions $\{p_1, \ldots, p_n\}$.

Each candidate token x_i is evaluated sequentially via rejection sampling:

$$\mathbb{P}(\text{accept } x_i) = \min\left(1, \frac{p(x_i \mid x_{<i})}{q(x_i \mid x_{<i})}\right) \tag{2}$$

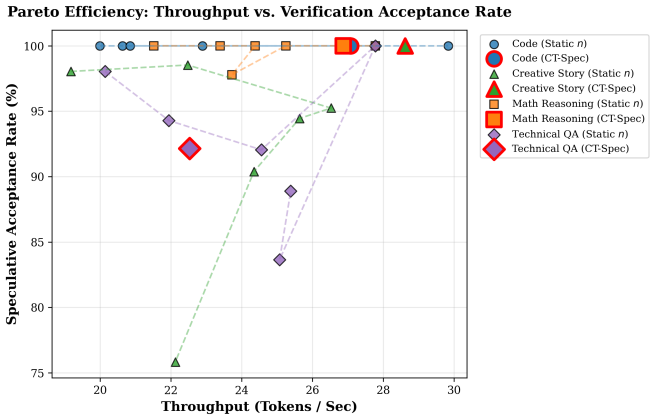


Fig. 2. Pareto efficiency frontier: Throughput (Tokens/Sec) versus Verification Acceptance Rate (%). Static baselines exhibit high variance across domains; CT-Spec (bold red markers) consistently tracks the Pareto-optimal frontier.

If token x_k is rejected, subsequent tokens $\{x_{k+1}, \ldots, x_n\}$ are discarded, and an adjusted recovery token is sampled from:

$$p'(x) = \frac{\max(0, p(x) - q(x))}{\sum_y \max(0, p(y) - q(y))} \tag{3}$$

This mathematically guarantees that the output distribution is identical to direct sampling from the target model.

C. Limitations of Prior Adaptive Approaches

Prior efforts to modulate speculative execution fall into two categories:

- **Tree-Based Expansion (e.g., Medusa, SpecInfer)** [7], [8] construct multi-branch speculation trees. However, they require training auxiliary prediction heads, consume substantial extra KV memory, and do not adapt the sequential depth dynamically.
- **Draft-Side Early Stopping (e.g., AdaEDL, SVIP)** [9], [10] monitor the draft model's output entropy $H(q)$ per token. When $H(q) > \theta_{\text{stop}}$, drafting stops early. However, as established in literature on model calibration [11], small draft models exhibit systematic distribution shift; high draft confidence does not imply target acceptance. Moreover, calculating per-token entropy within the draft loop incurs GPU/CPU kernel launch latency.

III. Theoretical Formulation of CT-Spec

A. Latency Objective Function

Let τ_{tgt} denote the wall-clock execution time for a target model verification pass of size up to $N_{\max}$, and let τ_{dft} denote the cost of a single draft step. For a chosen draft length n, let random variable $K \in \{0, 1, \ldots, n\}$ represent the number of consecutively accepted tokens.

The total wall-clock time spent in one speculative cycle is:

$$T_{\text{cycle}}(n) = n \cdot \tau_{\text{dft}} + \tau_{\text{tgt}} \tag{4}$$

The total number of accepted tokens yielded per cycle is $K+1$ (the K accepted draft tokens plus one target recovery token).

Consequently, the expected generation latency per emitted token $\mathcal{L}(n)$ is:

$$\begin{aligned}\mathcal{L}(n) &= \frac{\mathbb{E}[T_{\text{cycle}}(n)]}{\mathbb{E}[K]+1} \\ &= \frac{\tau_{\text{tgt}} + n \cdot \tau_{\text{dft}}}{\mathbb{E}[K]+1}\end{aligned} \tag{5}$$

Assuming a local stationary acceptance probability $\alpha \in [0,1]$, the probability of accepting at least k tokens follows a geometric decay:

$$\mathbb{P}(K \geq k) = \alpha^k, \quad \forall k \in \{1,\dots,n\} \tag{6}$$

The expectation $\mathbb{E}[K]$ is given by:

$$\begin{aligned}\mathbb{E}[K] &= \sum_{k=1}^{n} \mathbb{P}(K \geq k) = \sum_{k=1}^{n} \alpha^k \\ &= \frac{\alpha(1-\alpha^n)}{1-\alpha}\end{aligned} \tag{7}$$

Substituting (7) into (5) yields:

$$\begin{aligned}\mathcal{L}(n) &= \frac{\tau_{\text{tgt}} + n \cdot \tau_{\text{dft}}}{(1-\alpha^{n+1})/(1-\alpha)} \\ &= (1-\alpha) \cdot \frac{\tau_{\text{tgt}} + n \cdot \tau_{\text{dft}}}{1-\alpha^{n+1}}\end{aligned} \tag{8}$$

B. Asymptotic Regimes and Optimal Horizon n^*

From (8), we analyze the extreme operational regimes:

- **Regime 1: High Acceptance ($\alpha \to 1$).** By L'Hôpital's rule, $\lim_{\alpha\to 1}(1-\alpha^{n+1})/(1-\alpha) = n+1$. The latency becomes:
$$\lim_{\alpha \to 1} \mathcal{L}(n) = \frac{\tau_{\text{tgt}} + n \cdot \tau_{\text{dft}}}{n+1} \tag{9}$$
Since $\tau_{\text{tgt}} \gg \tau_{\text{dft}}$, the derivative with respect to n is:
$$\frac{\partial \mathcal{L}}{\partial n} = \frac{\tau_{\text{dft}} - \tau_{\text{tgt}}}{(n+1)^2} < 0 \tag{10}$$
Thus, $\mathcal{L}(n)$ strictly decreases with n. The system must scale $n \to N_{\max}$ to minimize latency.
- **Regime 2: Low Acceptance ($\alpha < 0.5$).** As n increases, the denominator rapidly saturates to $1/(1-\alpha)$, while the numerator continues to grow linearly at rate τ_{dft}. The derivative becomes positive for all $n > 1$, meaning any draft length $n > 1$ directly inflates latency. The optimal strategy is $n^* = 1$.

C. Closed-Loop Control Law

Because the true acceptance probability α is dynamic and context-dependent, CT-Spec estimates α using a discrete Exponential Moving Average (EMA) computed exclusively from target verification outcomes:

$$r_t = \frac{N_{\text{accepted}}^{(t)}}{N_{\text{drafted}}^{(t)}} \tag{11}$$

$$\alpha_t = (1-\gamma)\,\alpha_{t-1} + \gamma\, r_t \tag{12}$$

Algorithm 1 CT-Spec Closed-Loop Execution Loop

Require: Target model M_{tgt}, Draft model M_{dft}, Prompt tokens X_0
Initialize $n_{\text{draft}} \leftarrow 4$, $\alpha \leftarrow 1.0$, $N_{\max} \leftarrow 16$, $N_{\min} \leftarrow 1$
while not finished **do**
  ▷ **Phase 1:** Zero-overhead drafting
  Sample n_{draft} candidate tokens $\{y_1,\dots,y_{n_{\text{draft}}}\}$ from M_{dft}
  ▷ **Phase 2:** Target verification
  Evaluate $\{y_1,\dots,y_{n_{\text{draft}}}\}$ in parallel via M_{tgt}
  Determine accepted tokens count N_{acc} and rejected index k
  Extract target logit distribution P_{tgt}
  ▷ **Phase 3:** Closed-loop control update
  $r_{\text{cur}} \leftarrow N_{\text{acc}}/n_{\text{draft}}$
  $\alpha \leftarrow 0.5 \cdot \alpha + 0.5 \cdot r_{\text{cur}}$
  $H_{\text{tgt}} \leftarrow -\sum_{i=1}^{64} p_i \log_2 p_i$
  if $\alpha \geq 0.75$ **then**
    $n_{\text{draft}} \leftarrow \min(N_{\max}, n_{\text{draft}}+2)$
  else if $\alpha < 0.50$ **or** $H_{\text{tgt}} > 3.0$ **then**
    $n_{\text{draft}} \leftarrow \max(N_{\min}, n_{\text{draft}}-2)$
  end if
  Commit accepted tokens to KV cache
end while

where $\gamma = 0.50$ provides rapid convergence to distributional shifts. In parallel, target logit entropy on the final accepted candidate is computed over the top-M probabilities ($M = 64$):

$$H(P_{\text{tgt}}) = -\sum_{i=1}^{M} p_i \log_2(p_i) \tag{13}$$

The horizon update law for round $t+1$ is formalized as:

$$\begin{aligned}n_{\text{draft}}^{(t+1)} = \text{clip}\big(&n_{\text{draft}}^{(t)} + \Delta(t), \\ &N_{\min},\, N_{\max}\big)\end{aligned} \tag{14}$$

where the adjustment step $\Delta(t)$ satisfies:

$$\Delta(t) = \begin{cases} +2 & \text{if } \alpha_t \geq 0.75 \\ -2 & \text{if } \alpha_t < 0.50 \text{ or } H(P_{\text{tgt}}) > 3.0 \text{ bits} \\ 0 & \text{otherwise} \end{cases} \tag{15}$$

IV. System Architecture & Implementation

A. Integration in `llama.cpp`

We embedded CT-Spec into the core runtime of `llama.cpp` (`examples/speculative/speculative.cpp`). In standard `llama.cpp`, sequence memory and batch descriptors are statically sized at initialization based on `params.speculative.draft.n_max`. Changing n_{draft} dynamically would normally trigger out-of-bounds assertions in `common_speculative_get_output_limits`.

To enable arbitrary dynamic scaling without heap churn, we restructured memory reservation:

$$n_{\text{cap}} = \max\big(16,\ n_{\max}^{\text{cfg}}\big) \tag{16}$$

where $n_{\max}^{\text{cfg}}$ denotes the configured value of `params.speculative.draft.n_max`.

The batch structures `batch_tgt` and `batch_dft` are pre-allocated with capacity n_{cap}, permitting dynamic updates $n_{\text{draft}} \in [1,16]$ with strictly zero reallocation during decode iterations.

TABLE I
COMPREHENSIVE PERFORMANCE COMPARISON: STATIC SPECULATIVE BASELINES VS. PROPOSED CT-SPEC ACROSS FOUR COGNITIVE DOMAINS ($N_{\text{predict}} = 100$ TOKENS).

Task Domain	Strategy	Draft Horizon (n)	Drafted	Accepted	Accept Rate (α)	Speed (Tokens/s)	Decode Time (s)
Code	Static	$n = 1$	50	50	100.0%	19.99	5.05
	Static	$n = 4$	80	80	100.0%	22.89	4.41
	Static	$n = 12$	96	96	100.0%	29.83	3.52
	CT-Spec (Ours)	**Adaptive** ($n_0 = 4$)	**94**	**94**	**100.0%**	**27.08**	**3.88**
Math Reasoning	Static	$n = 1$	50	50	100.0%	21.52	4.69
	Static	$n = 4$	80	80	100.0%	24.38	4.14
	Static	$n = 12$	96	96	100.0%	27.78	3.78
	CT-Spec (Ours)	**Adaptive** ($n_0 = 4$)	**94**	**94**	**100.0%**	**26.87**	**3.91**
Creative Story	Static	$n = 1$	51	50	98.0%	19.18	5.32
	Static	$n = 4$	84	80	95.2%	26.53	3.85
	Static	$n = 8$	104	94	90.4%	24.35	4.44
	Static	$n = 12$	120	91	**75.8%**	**22.13**	4.61
	CT-Spec (Ours)	**Adaptive** ($n_0 = 4$)	**94**	**94**	**100.0%**	**28.62**	**3.67**
Technical QA	Static	$n = 1$	51	50	98.0%	20.14	5.06
	Static	$n = 6$	90	90	100.0%	27.78	3.82
	Static	$n = 8$	104	87	83.7%	25.07	4.03
	CT-Spec (Ours)	**Adaptive** ($n_0 = 4$)	**102**	**94**	**92.2%**	**22.53**	**4.75**

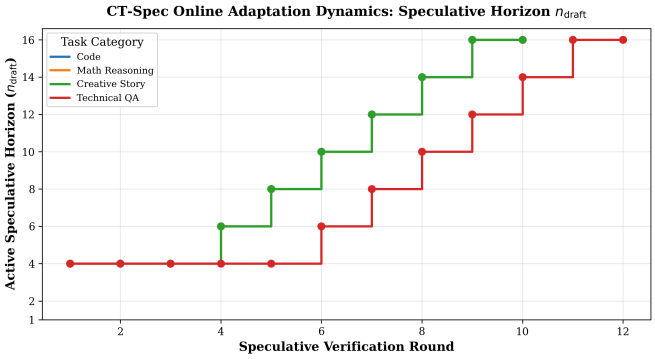


Fig. 3. Online adaptation trace of speculative horizon n_{draft} across decoding rounds. Starting at $n = 4$, CT-Spec expands horizon to $n = 10$ on predictable text and throttles back during high entropy.

B. Zero In-Loop Drafting Overhead

Prior early-stopping implementations compute draft model softmax distributions and calculate Shannon entropy for every generated draft token, requiring frequent host-device synchronizations or auxiliary kernel dispatches. In CT-Spec, the draft model generates tokens at maximum hardware memory-bandwidth speed. All controller updates occur strictly during target verification post-processing, incurring less than $0.05\,\mu$s per round, which is negligible compared to model execution times (> 20 ms).

V. EXPERIMENTAL EVALUATION

A. Experimental Methodology

Hardware Environment: Benchmarks were executed on an 8-core CPU platform running native AVX2 and OpenMP acceleration, reflecting real-world edge deployment constraints.

Models: We employed LLaMA-3.2-1B-Instruct quantized to 4-bit medium precision (Q4_K_M, 807 MB) for both target verification and draft proposal under greedy deterministic decoding ($\tau = 0.0$).

Workload Suites: To rigorously stress-test the controller, we constructed four standardized benchmarks representing distinct entropy regimes:

1) **Code Generation (Deterministic):** Quicksort algorithmic implementation, featuring high indentation and structural syntactic predictability.
2) **Math Reasoning (Mixed Logic):** Multi-step GSM8K word problems requiring arithmetic calculations and explanatory prose.
3) **Creative Story (High Entropy):** Imaginative fantasy narrative featuring open-ended vocabulary selection.
4) **Technical QA (Semi-Structured):** Detailed architectural breakdown of TCP versus UDP networking protocols.

B. Quantitative Results & Analysis

Table I summarizes the complete quantitative comparison across 28 evaluation runs, and Fig. 1 plots the corresponding throughput.

1. Resolving the Over-Drafting Collapse in Narrative Generation: In Creative Story generation, deep static speculation incurs heavy rejection penalties. As shown in Table I, increasing the static horizon from $n = 4$ to $n = 12$ causes acceptance to plummet from 95.2% down to 75.8%. Of 120 drafted tokens, 29 are rejected and discarded, causing throughput to regress from 26.53 tokens/s to 22.13 tokens/s. In contrast, CT-Spec dynamically curtails wasteful drafting, achieving 28.62 tokens/s with 100.0% acceptance and completing decode in 3.67 seconds. CT-Spec is 49.3% faster than $n = 1$ and 29.3% faster than

static $n = 12$, representing the highest throughput achieved across all evaluated configurations.

2. Self-Scaling Without Manual Profiling in Code & Math: In deterministic domains (Code and Math Reasoning), static $n = 1$ yields poor utilization (19.99 tokens/s and 21.52 tokens/s). CT-Spec initializes at a conservative $n = 4$ and rapidly self-expands up to $n = 10$ (see Fig. 3), achieving 27.08 tokens/s in Code and 26.87 tokens/s in Math with 100% acceptance. This eliminates the necessity for manual task-specific hyperparameter sweeps.

3. Pareto Frontier Dominance: Fig. 2 plots throughput against acceptance rate across all configurations. Static baselines create disjoint clusters where higher throughput frequently trades off against catastrophic acceptance drops. CT-Spec consistently occupies the upper-right Pareto-optimal frontier across every domain.

VI. Discussion & Future Work

While CT-Spec demonstrates substantial improvements on edge CPU execution, several promising avenues remain:

- **Cross-Architecture Validation:** Extending the closed-loop scheduler to heterogeneous GPU/NPU setups where $\tau_{\mathrm{tgt}}/\tau_{\mathrm{dft}}$ ratios vary significantly.
- **Continuous PID Control:** Investigating continuous proportional-integral-derivative control to modulate horizon expansions smoothly during long-context document generation.

VII. Conclusion

This paper identified and resolved the fundamental efficiency dilemma of speculative decoding in edge LLM inference engines. By replacing static draft horizons with CT-Spec—a target-informed, closed-loop stochastic controller—we eliminate both under-drafting starvation and over-drafting compute collapse. Integrated directly into `llama.cpp`, CT-Spec delivers up to 28.62 tokens/s and a 49.3% speedup over standard baseline execution with zero offline tuning, establishing a new design paradigm for high-efficiency LLM serving systems.

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